



Medical Microbiology

Lecture Sixteen

Mycology

Third stage- College of Dentistry

Dr. Dheyaa Naji Hamza

Dheyaa.naji@alkafeel.edu.iq

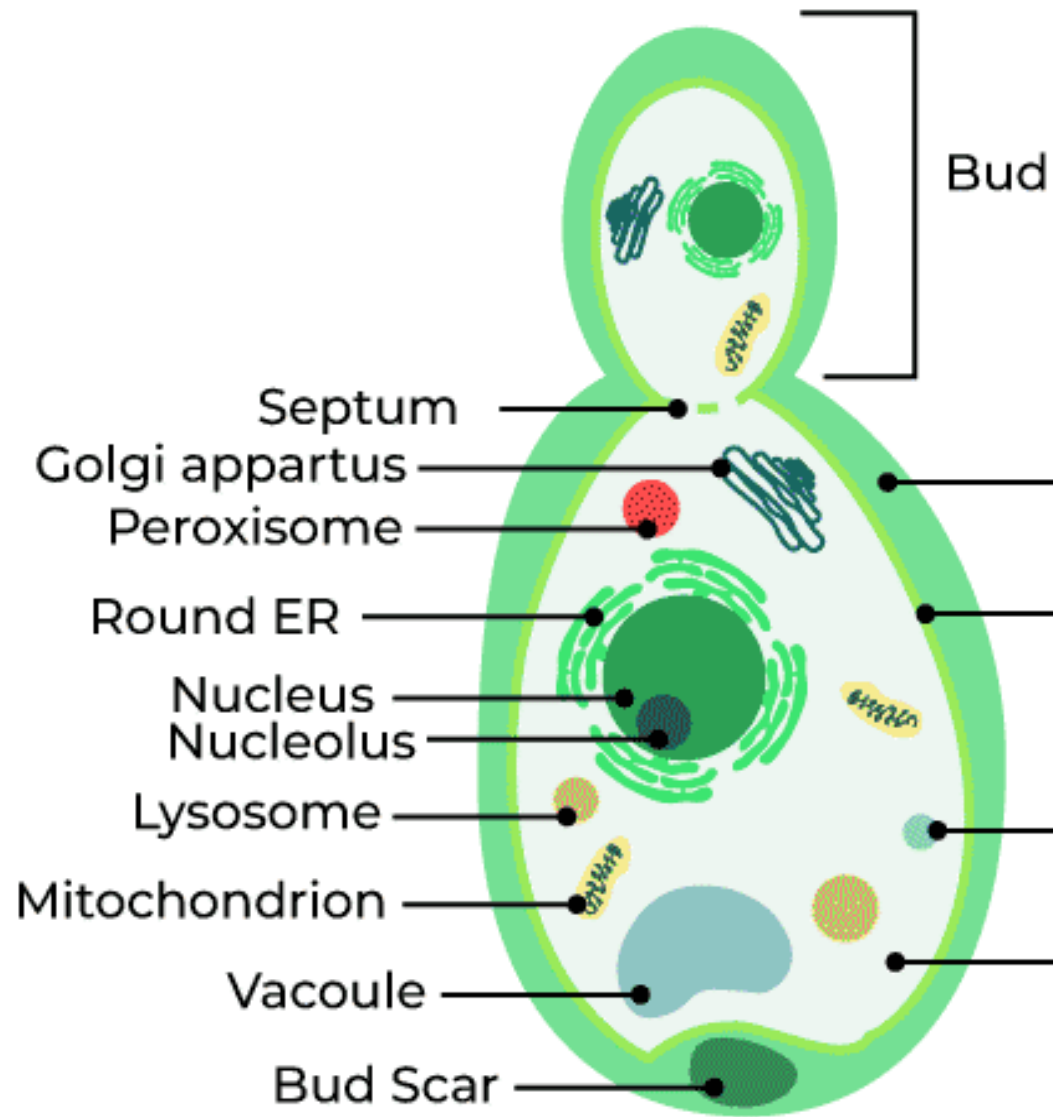
Introduction:

- ❖ **Mycology** is the study of fungi.
- ❖ **Fungi** are eukaryotic microorganisms that can exist as unicellular (yeasts) or multicellular (molds) organisms.
- ❖ They are widely distributed in nature and play important roles in both health and disease.
- ❖ They differ from the bacteria, which are prokaryotic organisms.

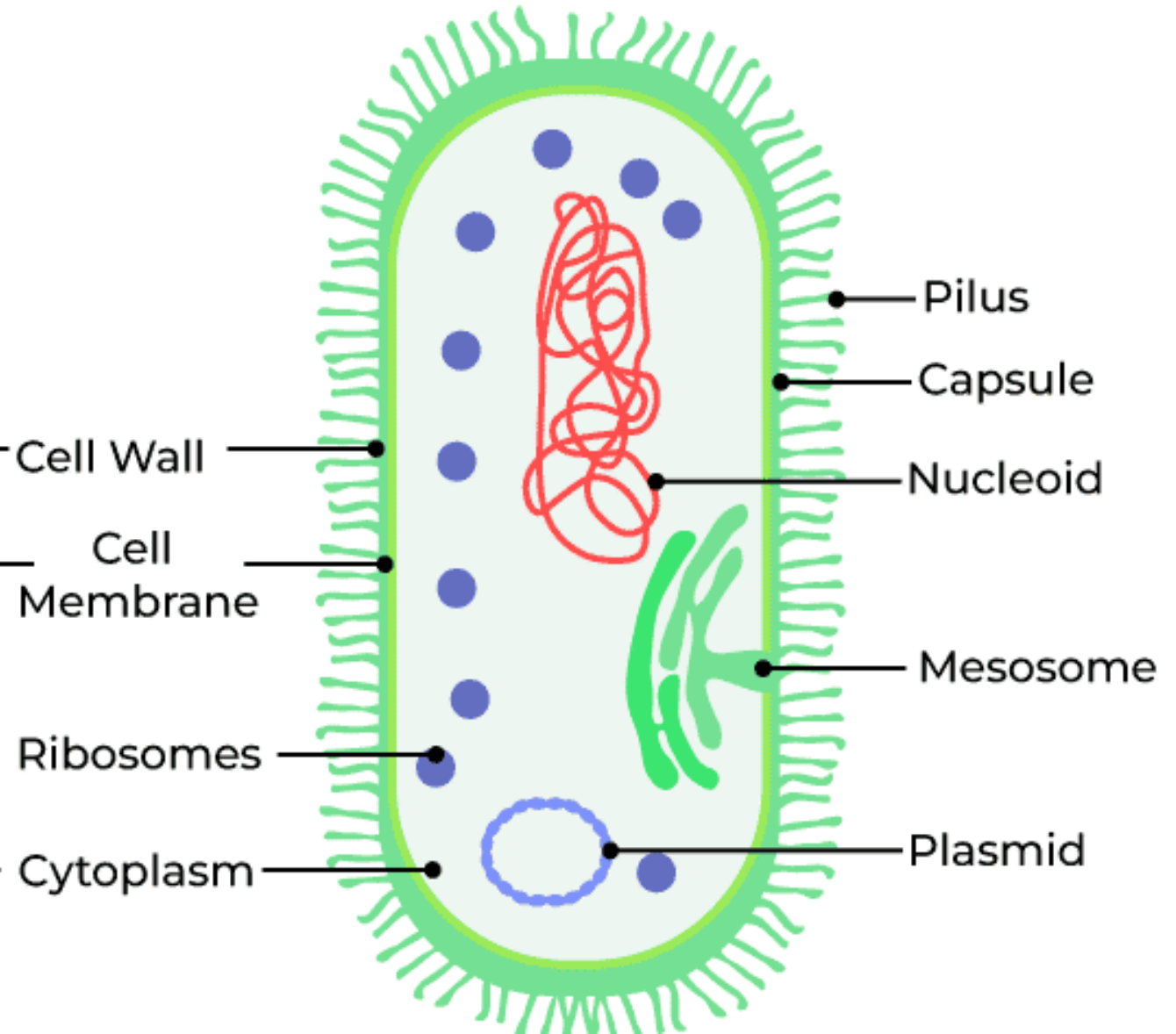
Yeasts **VS** Molds



Fungal Cell



Bacterial Cell



General Characteristics of Fungi:

- 1-Eukaryotic cells (true nucleus).
- 2-Cell wall contains chitin and glucan.
- 3-Cell membrane contains ergosterol (important drug target).
- 4- Fungi can **reproduce** sexually by forming sexual spores and asexually by forming conidia or asexual spores.
- 5- **Can be:**
 - **Yeasts** (unicellular, round/oval).
 - **Molds** (filamentous, hyphae).
 - **Dimorphic** fungi (exist as yeast or mold depending on temperature of environment).

Classification of Fungi (Medical Importance):

1-Yeasts: are round or oval unicellular fungi that reproduce by asexual budding.

- Example: *Cryptococcus neoformans*

2-Yeast-like fungi: These are the yeasts with pseudo hyphae.

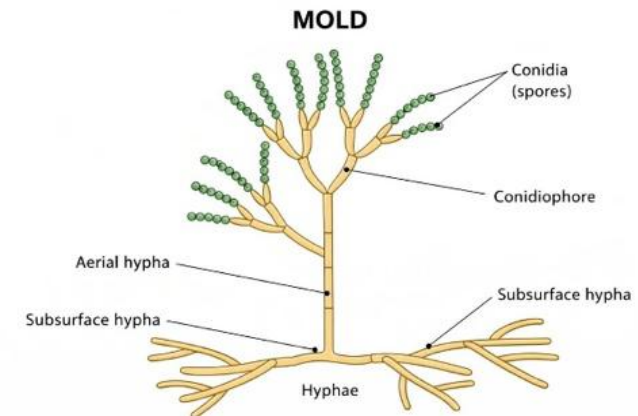
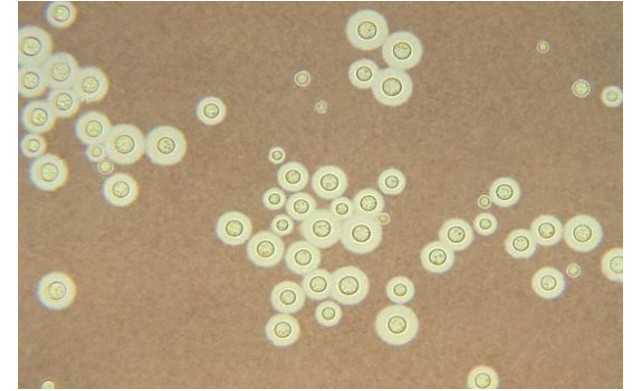
- Example: *Candida albicans*.

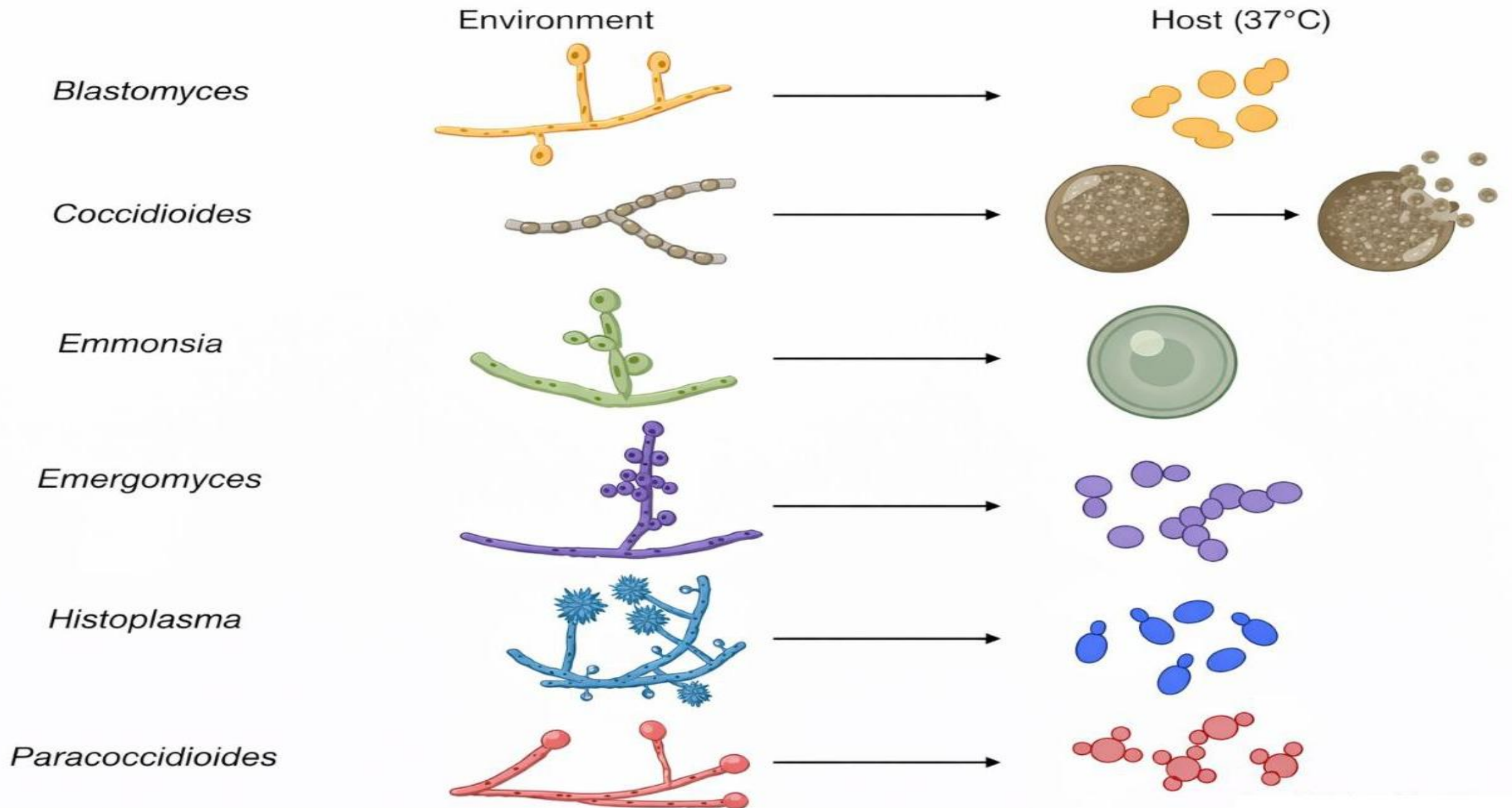
3- Molds:

- Example: *Aspergillus*, *penicillium*, etc..
- Form hyphae and mycelium.

4- Dimorphic Fungi: Many of medically important fungi are dimorphic.

- Example: *Histoplasma capsulatum*.
- Mold in environment, yeast in human body.



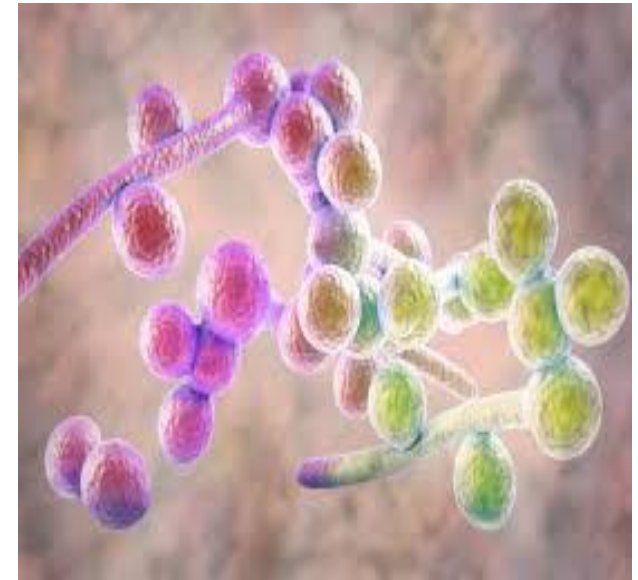


Candida albicans:

❖ **Candida albicans** is the most important opportunistic fungal pathogen in humans and is part of the normal flora of the oral cavity, gastrointestinal tract, and vagina.

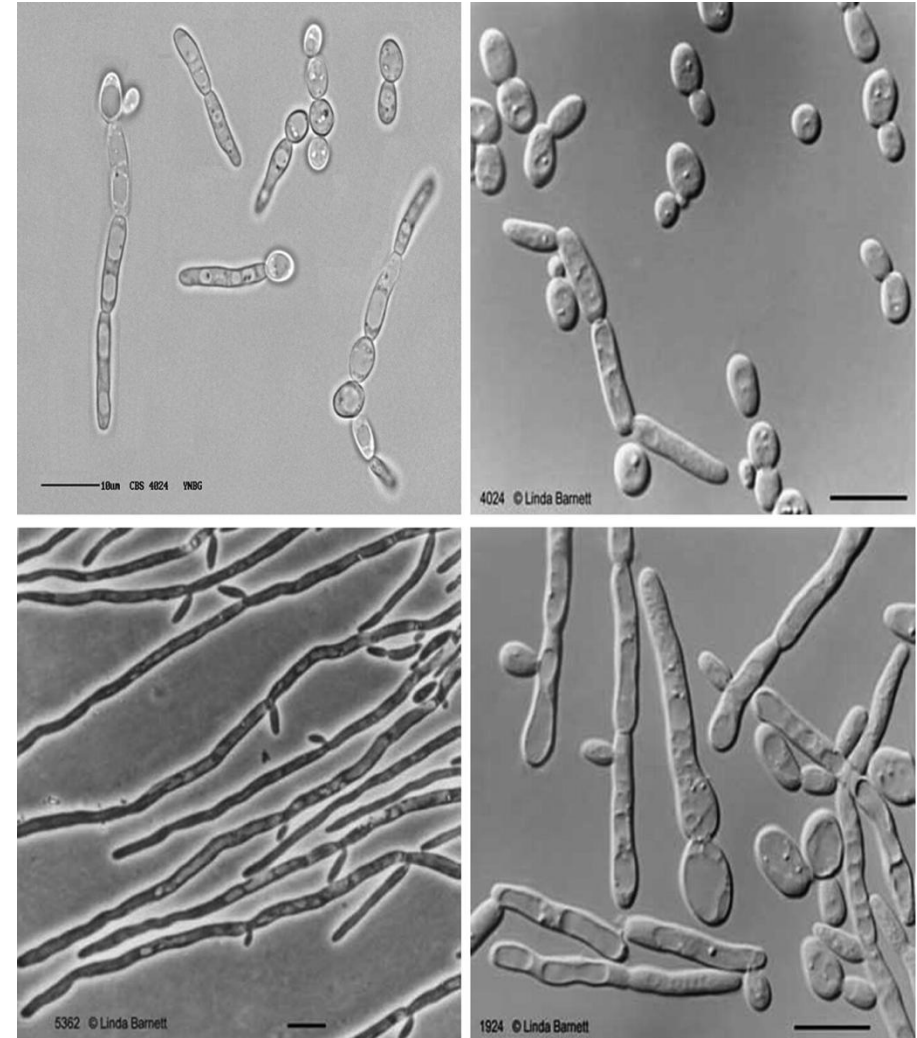
Morphology:

- Yeast form (budding cells).
- Pseudo hyphae (elongated chains of cells).
- True hyphae (invasive form).



Virulence Factors:

- 1-Adhesion to epithelial cells.
- 2-Biofilm formation (important in dentures).
- 3-Enzymes (proteases, phospholipases).
- 4-Dimorphism (yeast $\leftrightarrow$ hyphae transition).



Pathogenesis of Candida Infection:

Candida is normally harmless but becomes pathogenic when host defenses are compromised.

Risk Factors :

- Immunosuppression (HIV, chemotherapy).
- Diabetes mellitus.
- Broad-spectrum antibiotics.
- Poor oral hygiene.
- Denture use.
- Xerostomia (dry mouth).

Oral Candidiasis (Clinical Forms):

1. Pseudomembranous Candidiasis (Thrush):

- ☐ White curd-like plaques.
- ☐ Can be wiped off leaving red surface.

2. Erythematous (Atrophic) Candidiasis:

- ☐ Red, painful mucosa.
- ☐ Common after antibiotic use.

3. Chronic Hyperplastic Candidiasis:

- ☐ White lesions that cannot be removed.
- ☐ May have malignant potential.



4. Denture Stomatitis:

- ❑ Red inflammation under dentures.
- ❑ Strongly associated with biofilm.



5. Angular Cheilitis:

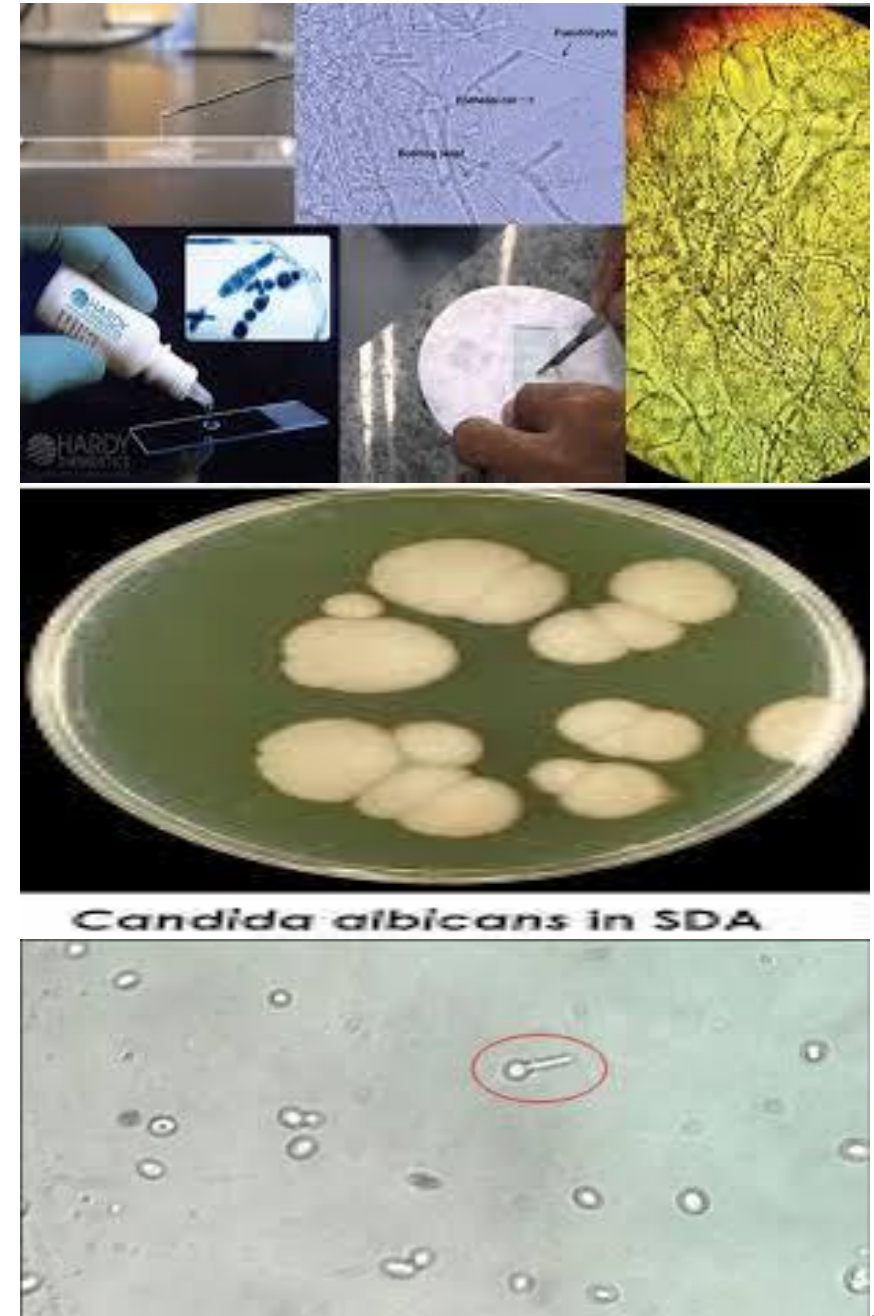
- ❑ Cracks at corners of mouth.
- ❑ Often mixed infection (Candida + bacteria).



Diagnosis of Candida:

Laboratory Methods:

- 1-Microscopy (KOH preparation).
- 2-Gram stain (Gram-positive yeast).
- 3-Culture on Sabouraud Dextrose Agar.
- 4-Germ tube test (specific for *Candida albicans*).



Candida albicans in SDA

Treatment of Oral Candidiasis:

1- Topical Antifungals:

- Nystatin.
- Clotrimazole.

2- Systemic Antifungals:

- Fluconazole.
- Itraconazole.

3- Management in Dentistry:

- Improve oral hygiene.
- Proper denture cleaning and removal at night.
- Control systemic diseases.



Importance in Dentistry:

❖ Candida albicans is highly relevant in dental practice because:

- 1-Common cause of oral infections.
- 2-Associated with dentures (biofilm formation).
- 3-Indicator of systemic disease (e.g., HIV).
- 4-May complicate antibiotic therapy.

Role of the Dentist:

- 1-Early detection of oral lesions.
- 2-Patient education.
- 3-Proper denture management.
- 4-Referral when systemic disease is suspected.



Summary:

- Fungi are important opportunistic pathogens.
- *Candida albicans* is the most common oral fungal infection.
- Oral candidiasis presents in different clinical forms.
- Dentists play a key role in diagnosis, treatment, and prevention.



Any Question?



Thank you